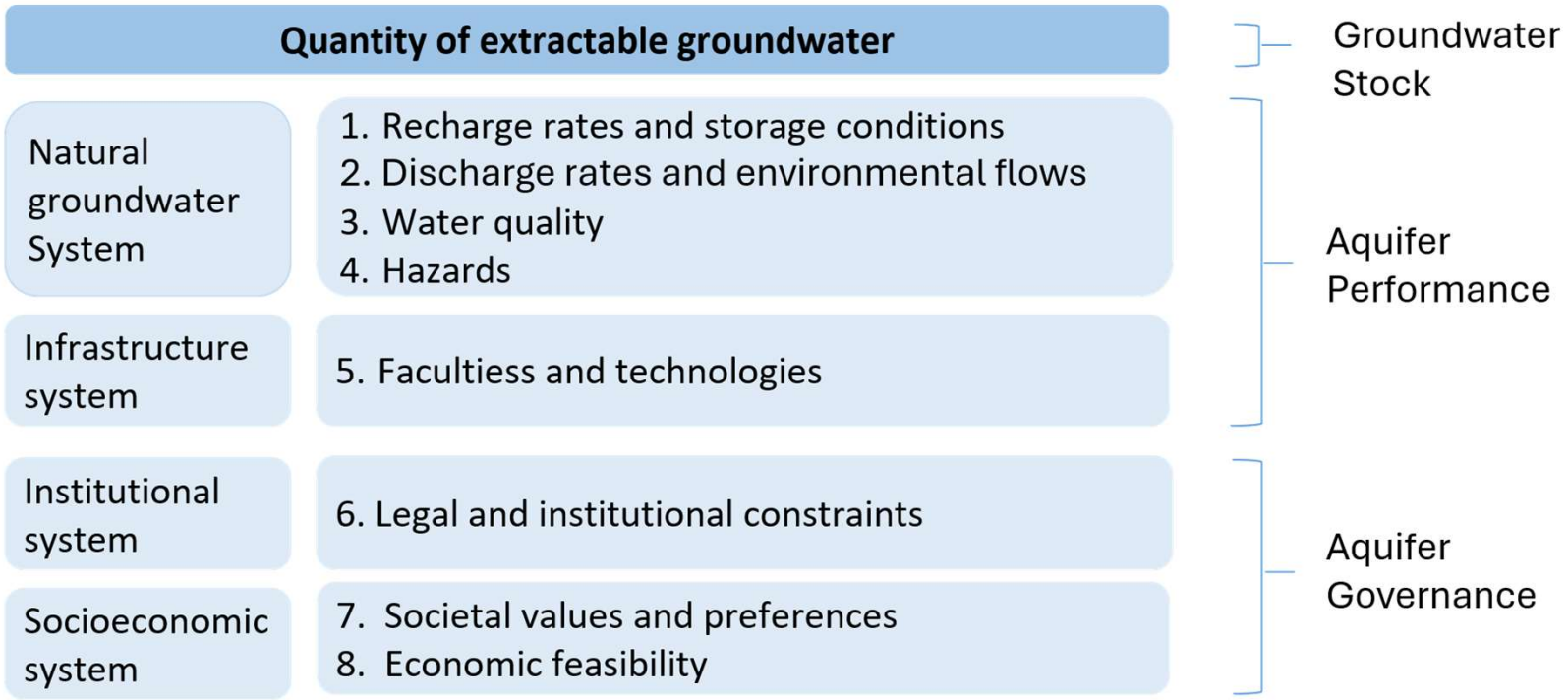
Florida Gulf Coast University

Groundwater Sustainability in South Florida

Ahmed S. Elshall Mewcha A. Gebremedhin Michael Alvear

November 9, 2026

Groundwater sustainability involves eight pillars related to aquifer performance and governance



Pierce et al. 2013 and Elshall et al., 2022

Nature-based Infrastructure for Enhancing Climate Resiliency of Groundwater Resources in South Florida


Ebrahim Ahmadisharaf, FSU


Ahmed Elshall, WCE and TWS


Mewcha Gebremedhin, WCE


Shelly Krueger, UF


Rachel Rotz, TWS


Seneshaw Tsegaye, WCE


Ming Ye, FSU

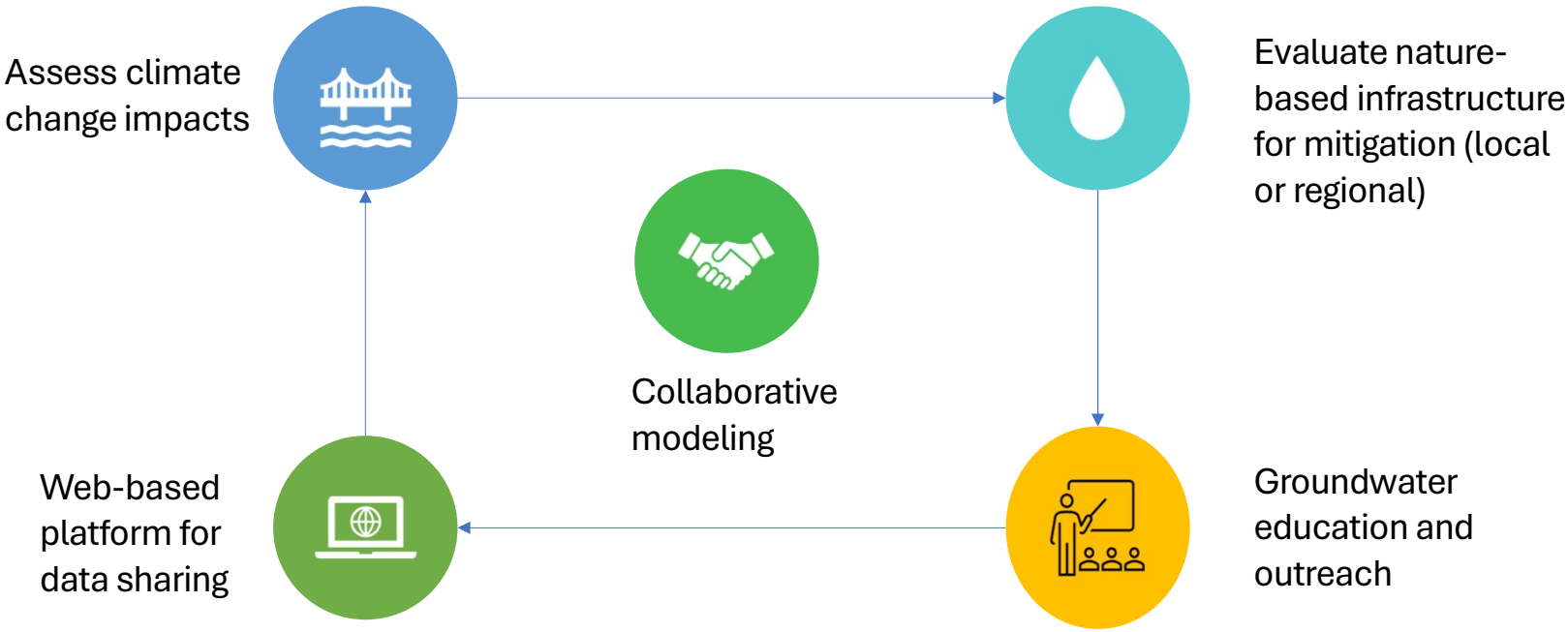
USGS science for a changing world

USGS Caribbean-Florida Water Science Center

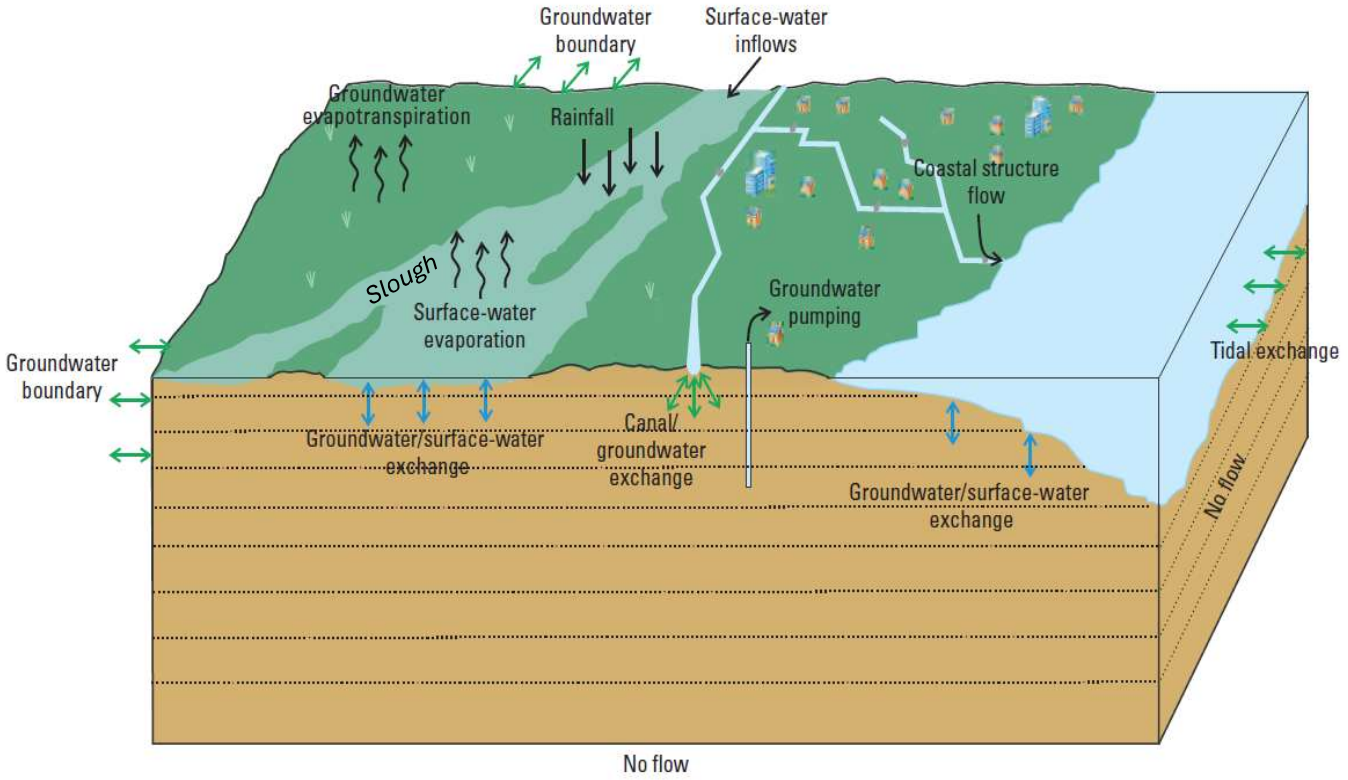
SOUTH FLORIDA WATER MANAGEMENT DISTRICT
PROTECTOR OF THE EVERGLADES SINCE 1949

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY

Project Overview

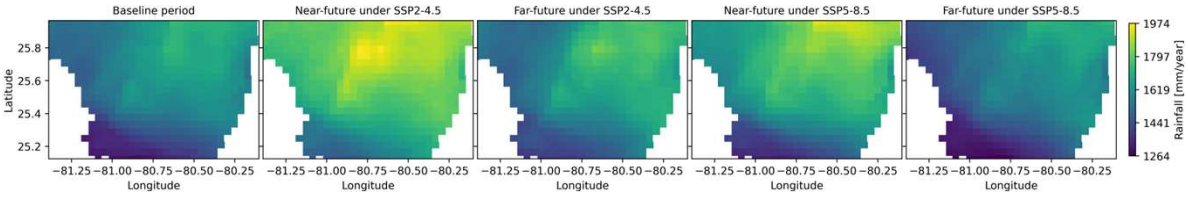


Climate change impacts on groundwater-surface water simulated using numerical models

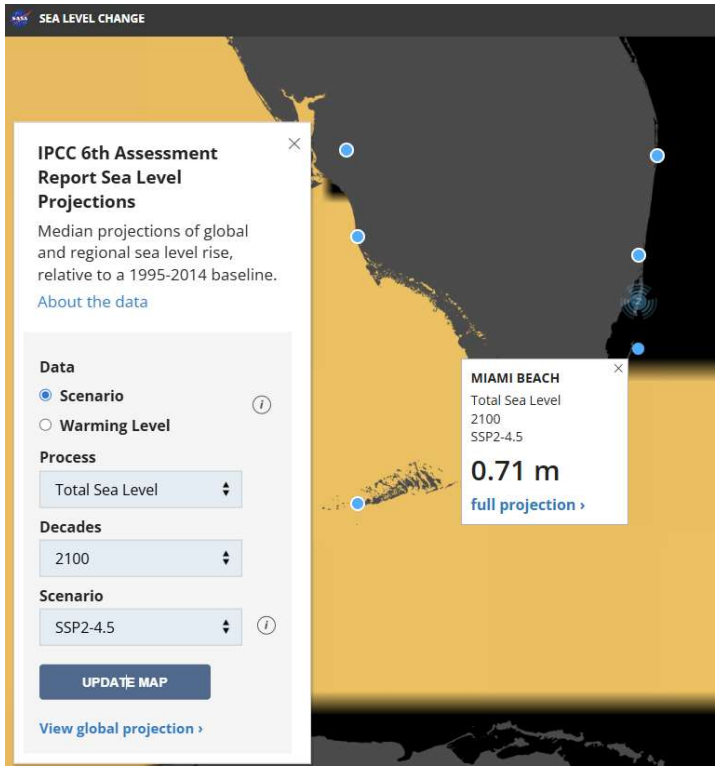


Biscayne and Southern Everglades Coastal Transport (BISECT) model (Swain, et al. 2019, USGS)

Model Inputs include climate projections with baseline and future temporal coverage



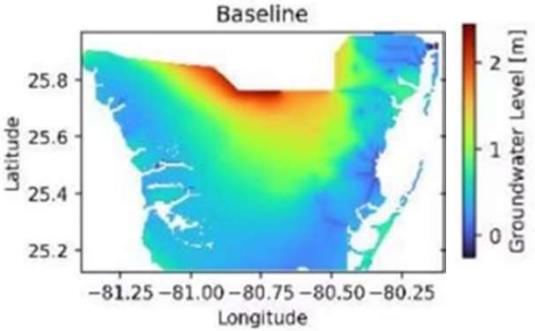
Rainfall, CMIP6 - Oak Ridge National Laboratory



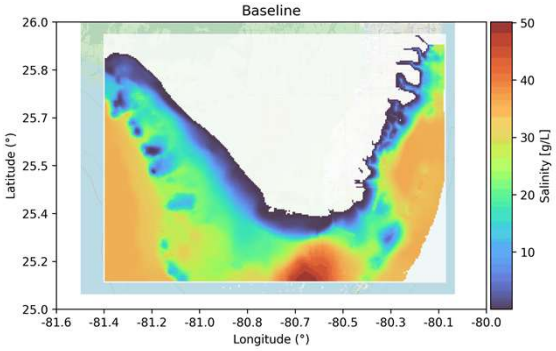
Sea-level Rise, CMIP6 - NASA

Gebermedhin et al. (2026)

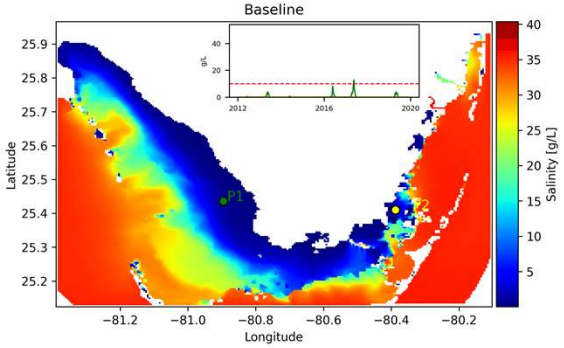
Model outputs are critical for diverse stakeholders at South Florida



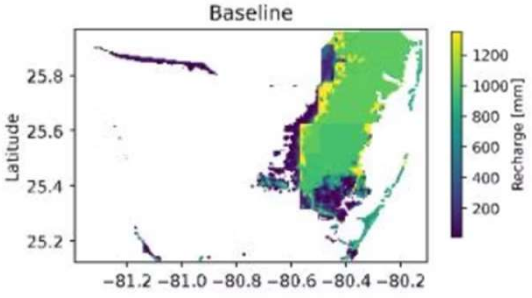
Groundwater Levels



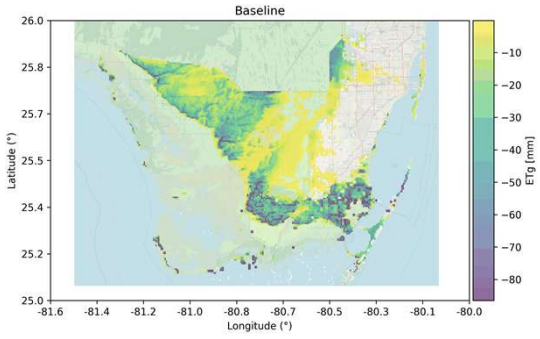
Groundwater Salinity



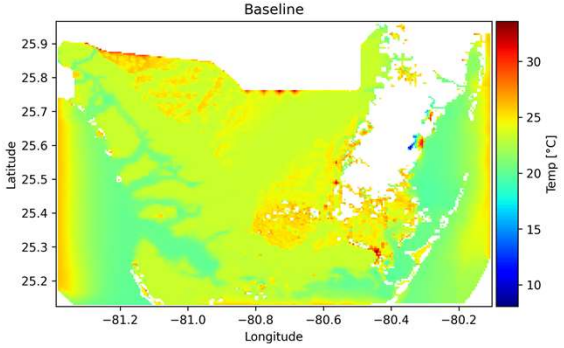
Surface-water Salinity



Groundwater Recharge



Groundwater Evapotranspiration



Surface-water Temperature

Gebermedhin et al. (2026)

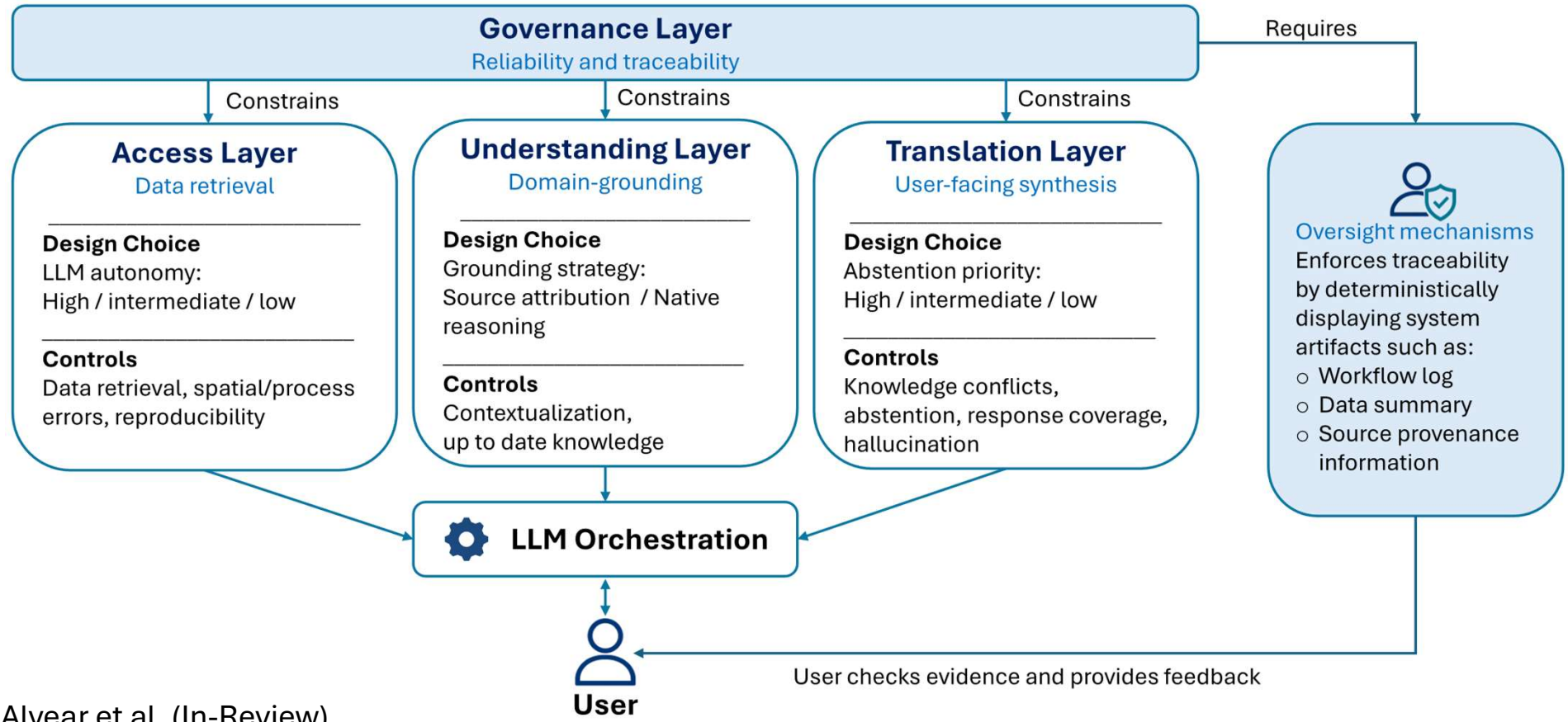
Making Hydrological information accessible through LLMs



Alvear et al. (In-Review)

https://clio-552796691518.us-east1.run.app/

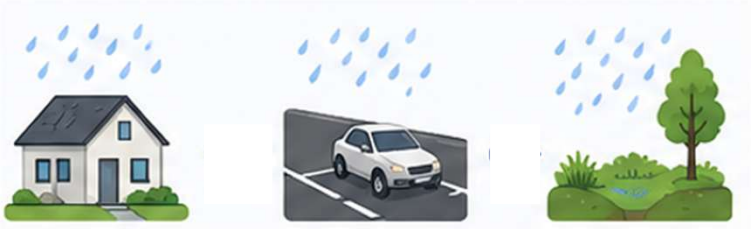
AI Governance for reliable and traceable environmental LLM systems



Alvear et al. (In-Review)

Green infrastructure for mitigation of increased stormwater runoff from urban development

Runoff Sources



Roof Parking/Road Pervious Surface

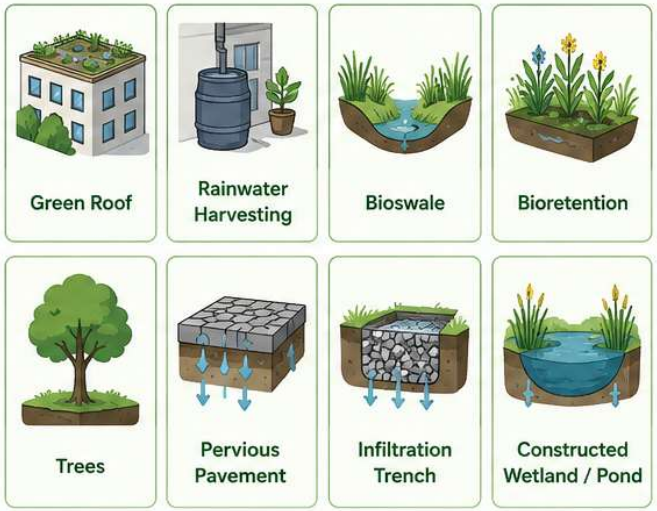


Gray infrastructure

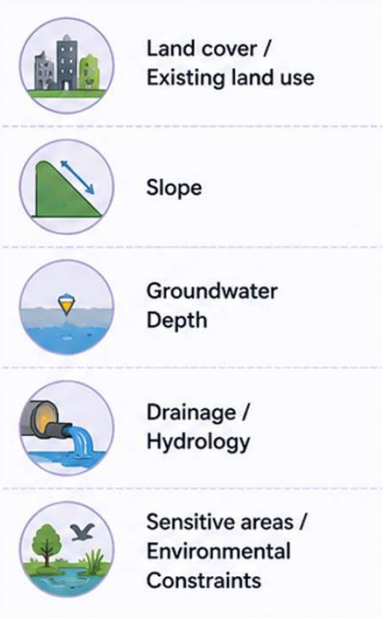


Green infrastructure

Green Infrastructure (BMPs)



Site conditions



GI PLACE: Geospatial method for BMPs placement

ChatGPT generated icons

1. Existing conditions



Data Layers

- Land cover
- Terrain
- Soil
- Buildings footprints
- Hydrological condition
- Existing green infrastructure

Existing hydrology and green infrastructure are mapped

• • •

2. Binary feasibility



Feasibility constraints

- Location requirement
- Proximity to impervious source
- Terrian threshold
- Groundwater separation
- Required buffers

Feasible

- Yes
- No

Only locations satisfying feasibility constraints are retained

• • •

3. Multicriteria prioritization



Priority criteria

- Runoff generation
- Hydraulic conductivity
- Storage adequacy
- Drainage proximity
- Terrain suitability

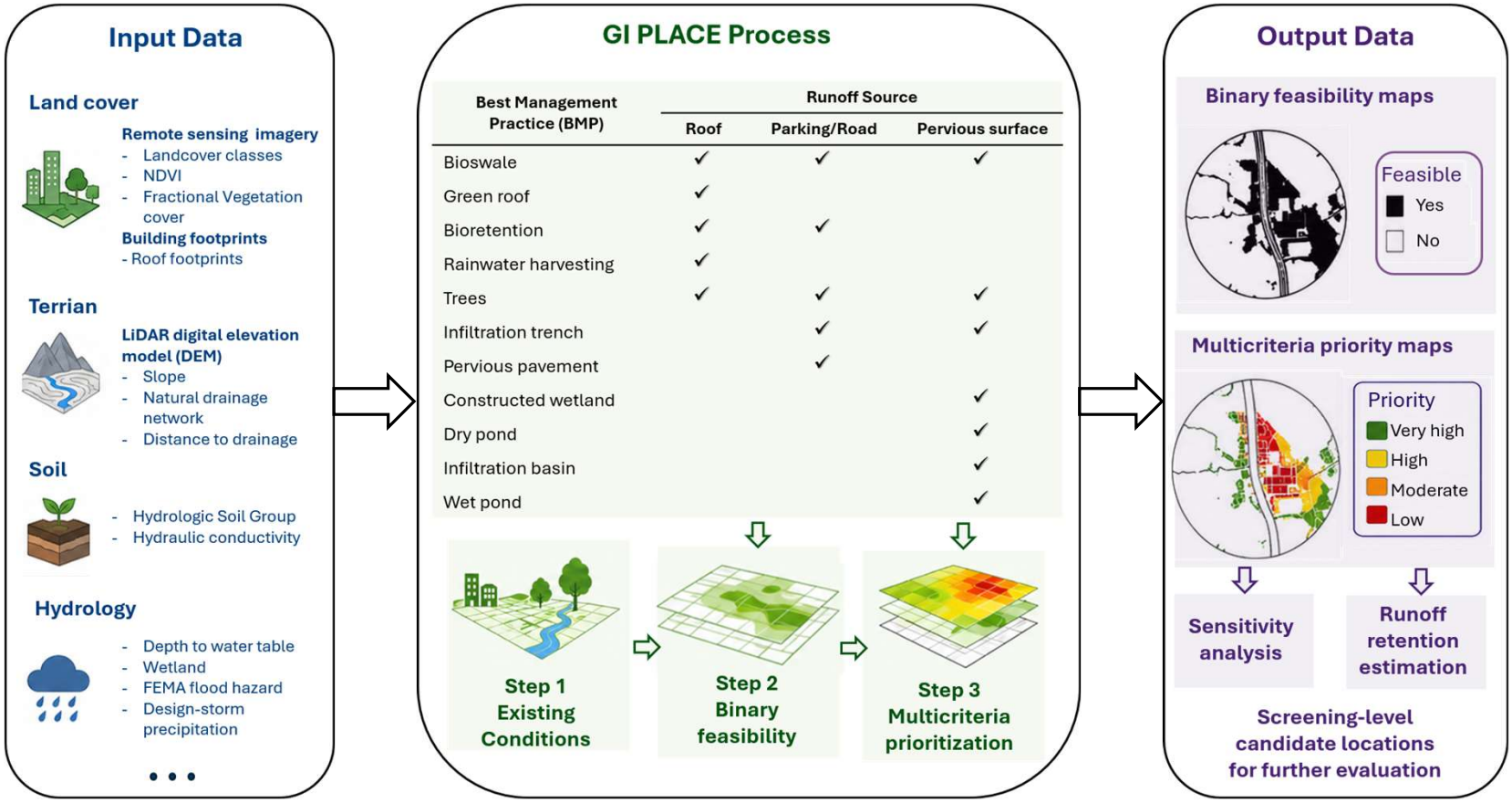
Priority

- Very high
- High
- Moderate
- Low

Only feasible locations are ranked

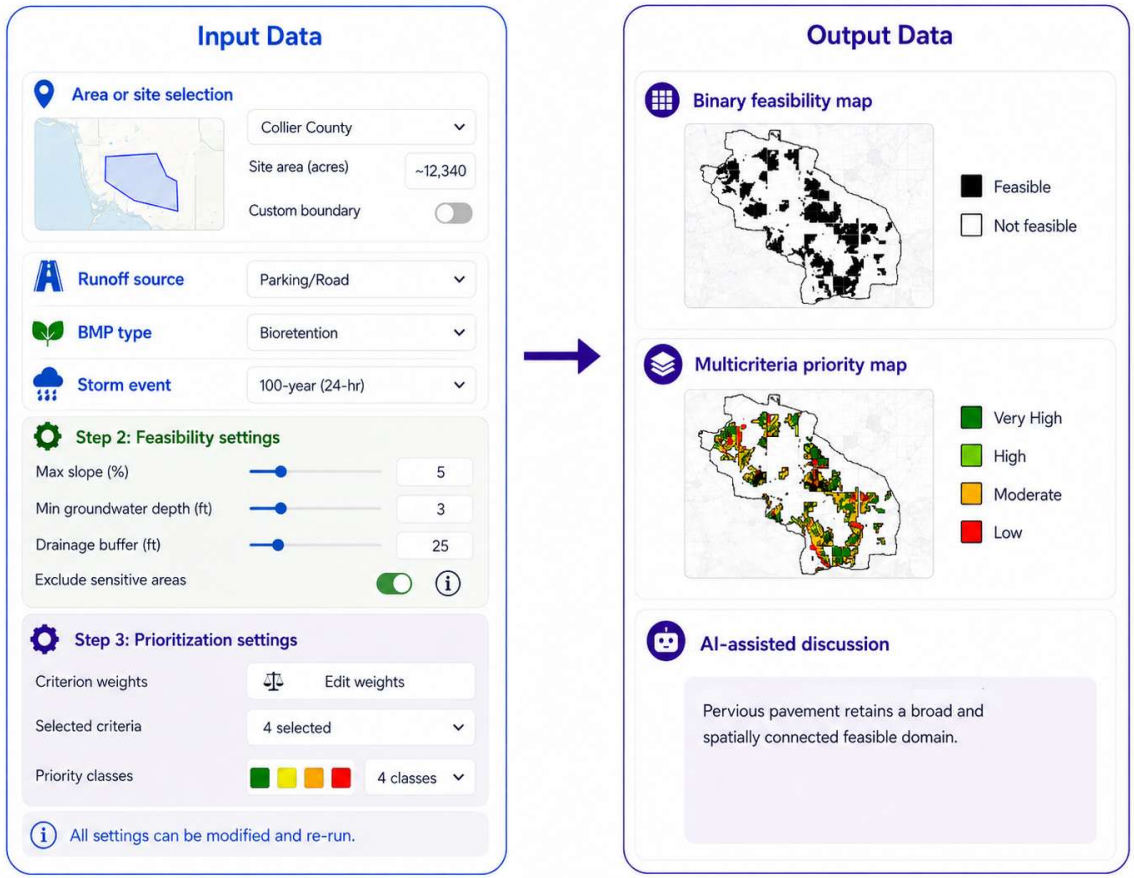
• • •

GI PLACE to interactive dashboard for GI Screening



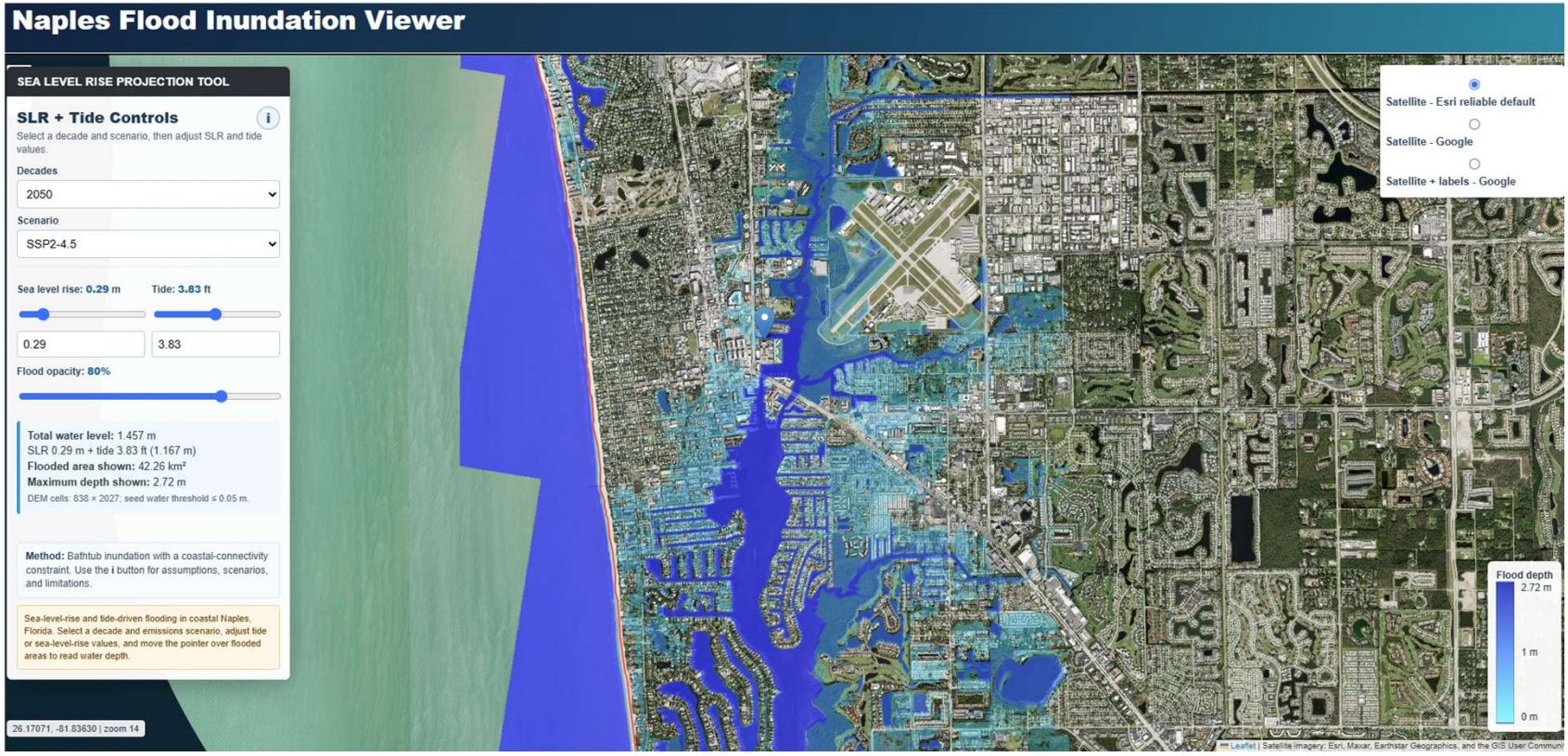
Gebermedhin et al. (In-Review)

AI-enabled dashboard: User selects, GI PLACE calculates, the dashboard shows and AI explains



Instructed ChatGPT generated image

Coastal flood screening in South Florida: Inundation model and data-driven rainfall–runoff model



Humic Acid for Nutrient Reduction and Red Tide Mitigation



Ahmed Elshall
WCE and TWS



Mewcha Gebremedhin
WCE



Suraj Melkani
UF/IFAS



Seneshaw Tsegaye
WCE



Nikolaos Tziolas
UF/IFAS



Ming Ye
FSU

FGCU

UF UNIVERSITY of FLORIDA

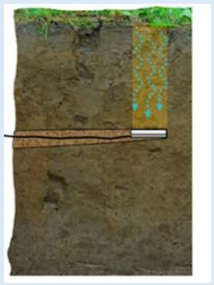
FSU

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY

Humic Acid for reducing nutrient loads and HAB risks

Task 1: Lysimeter-Scale Experiment

Quantify Nutrient Leaching to Demonstrate Improvement in Groundwater – Surface Water Quality



Suction plate lysimeter (ecoTech)

Task 2 Field-Scale Validation

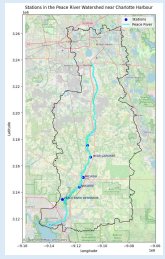
Quantify Crop Yield and Quality to Demonstrate Improvement in Farm Economy while Improving Water Quality

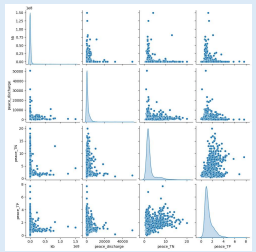


Tomato field, UF-IFAS, Immokalee

Task 3: Watershed-Scale Modeling

Quantify Gulf Nutrient Load Reduction and Demonstrate Decrease in Red Tide Risk





Higher nutrient loads a Peace River watershed correspond to increased bloom intensity

Technological Innovations

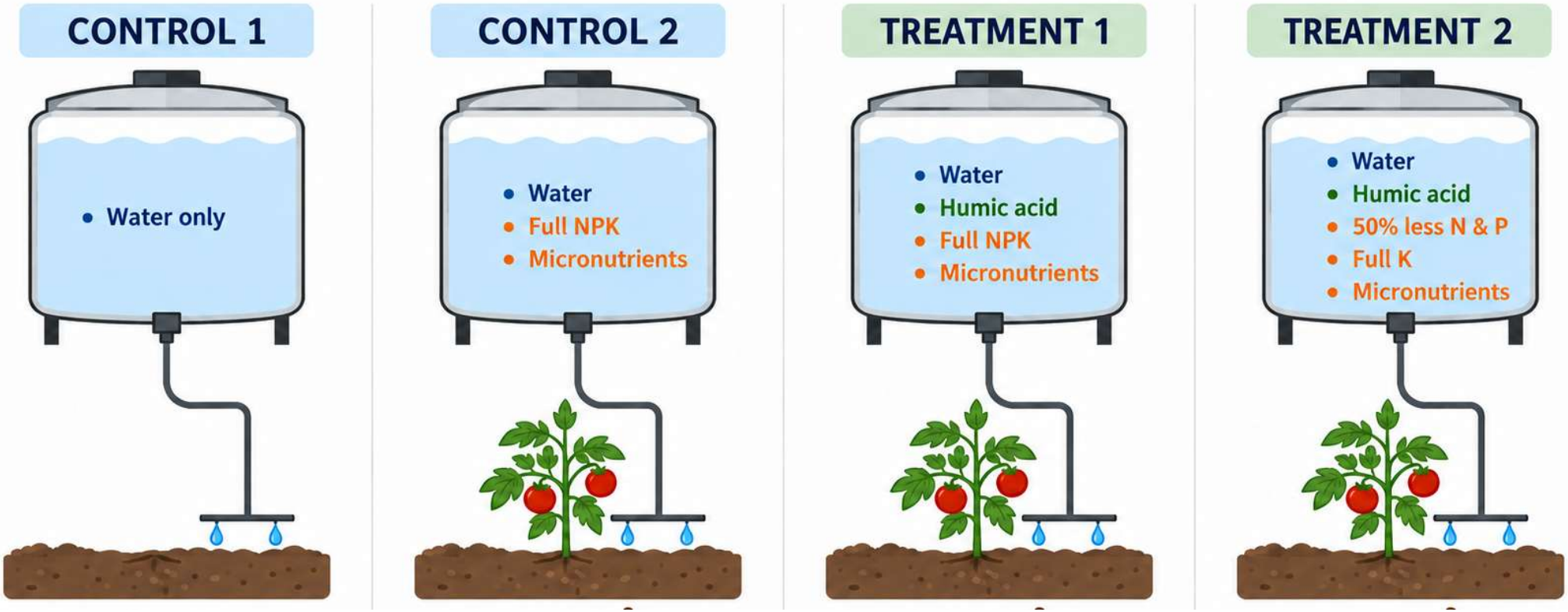
(A) Humic Acid to Minimize Nutrient Leaching and Improve Farm Economy

(B) UAV to Document Crop Heath and Support Stakeholder Engagement

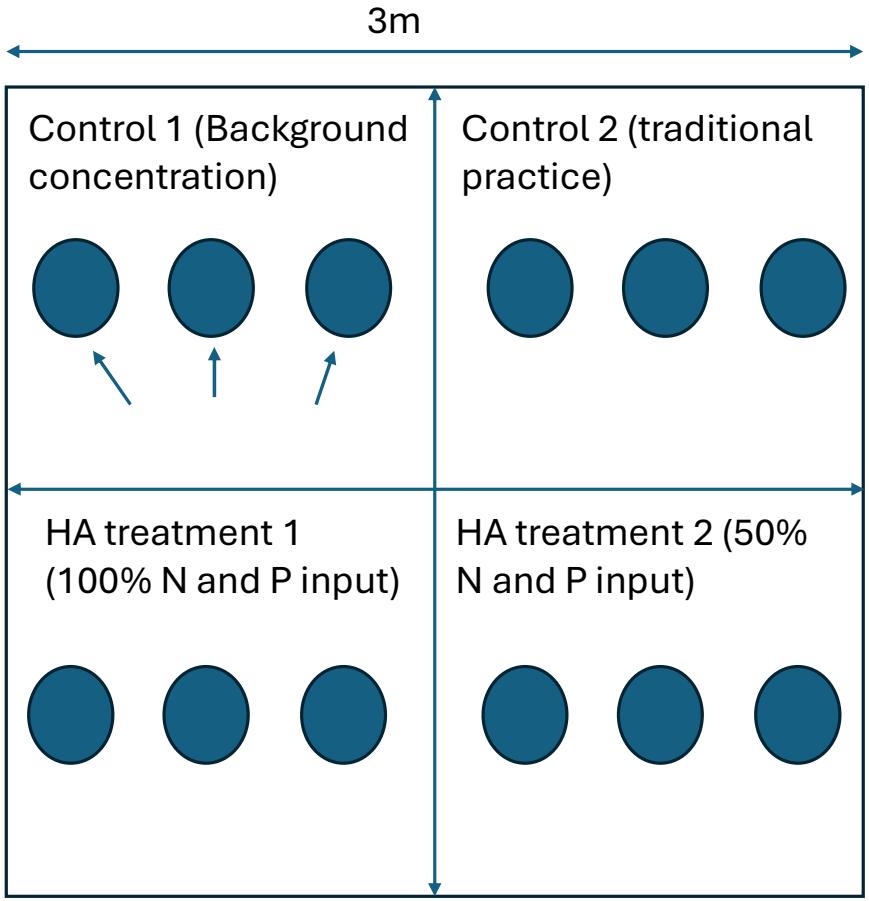
(C) Machine Learning Model to Predict Red Tide Mitigation given Agricultural Best Management Practices with Humic Acid

(D) AI-LLM Powered Decision-Support Dashboard for Agricultural Stakeholders

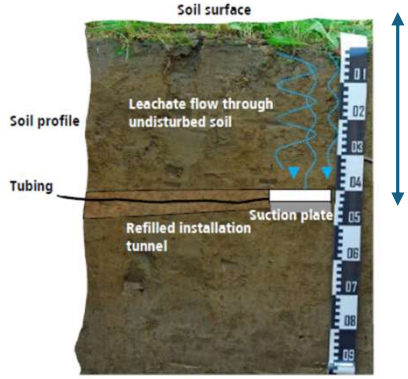
Task 1: Demonstrate at lysimeter scale the role humic acid in reducing leaching and improving water quality



Task 1: Demonstrate at lysimeter scale the role humic acid in reducing leaching and improving water quality



Find root depth and stay below root zone 5 cm – 10 cm



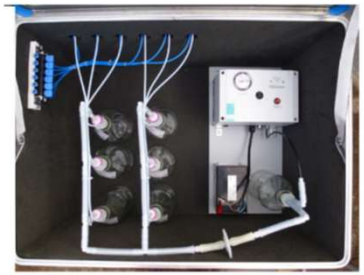




Suction plates







Sampling station

Lysimeter experiment to quantify TN and TP leaching to groundwater (3 replicates)

Task 2 Demonstrate at field scale the efficacy of humic acid in improving crop quality and reducing cost



Tomato field, UF-IFAS, Immokalee



Soil health and water dynamics



Crop health monitoring



Crop yield and economic analysis

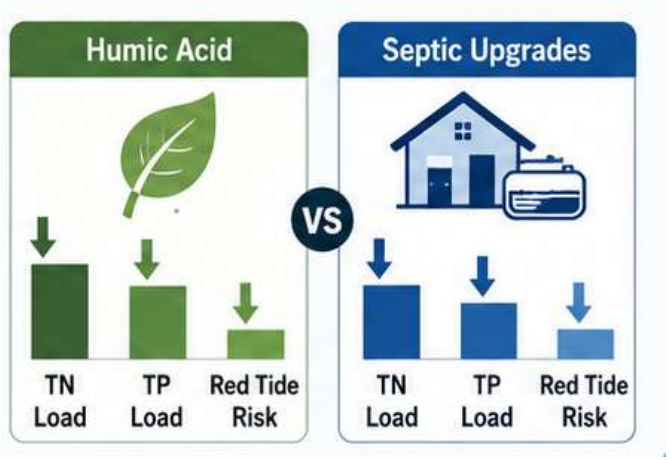


AI-enabled dashboard for growers

Task 3 Demonstrate at watershed scale reduction in nutrient input to Gulf water and red tide risk



Watershed upscaling



Scenario Analysis

Advance groundwater sustainability through observation, modeling, hydroinformatics, and participatory governance

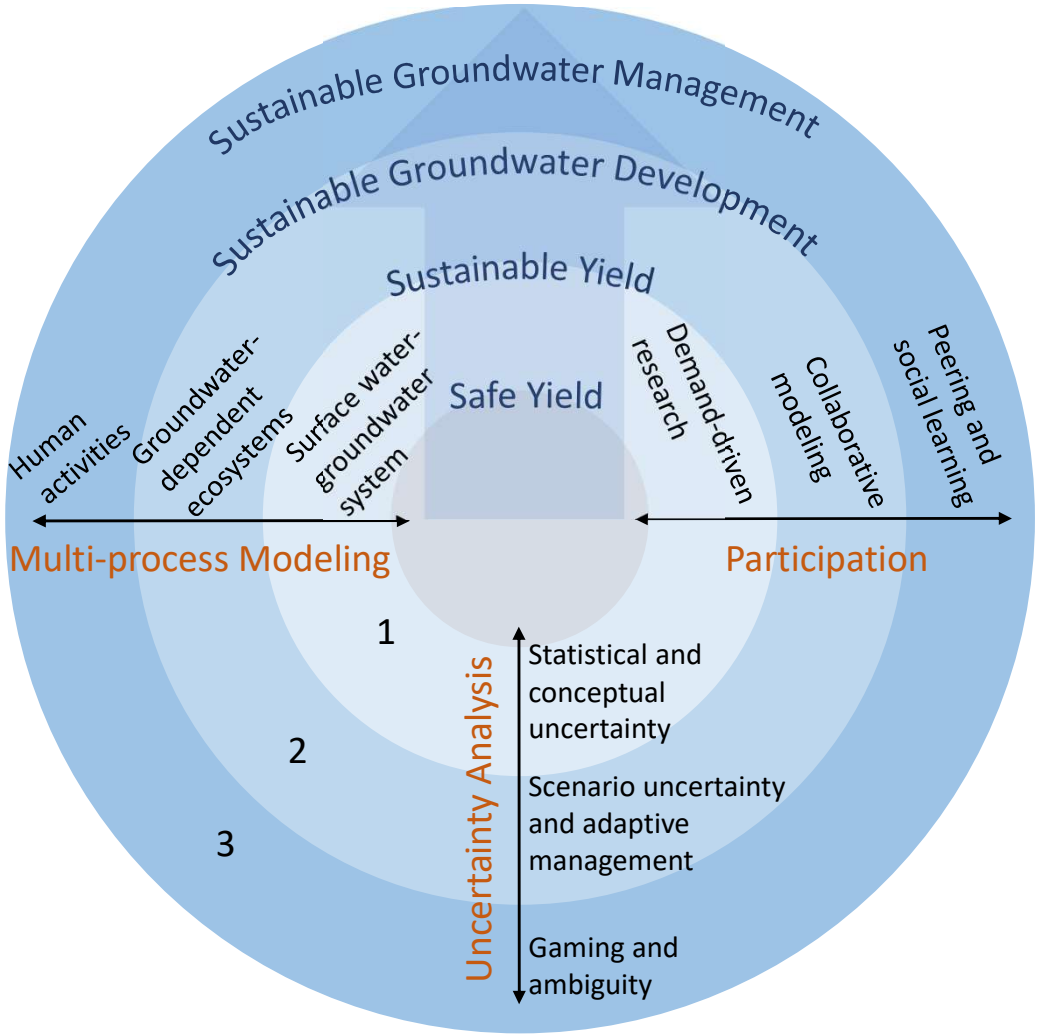
1. Integrated hydrological modeling
2. Geoinformatics for decision support
3. Participatory governance

Florida Gulf Coast University

Surface Water-Groundwater Lab

https://aselshall.github.io/gwh/lab

Groundwater sustainability is deeply participatory



Elshall et al. (2020)